

David Liu

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EDUCATION

Bachelor of Science (BS), Engineering

Harvey Mudd College • Claremont, CA, USA • 2028 • 3.741

EXPERIENCE

Electro-Mechanical Engineering Intern

MRP Solutions

May 2026 – Present, Claremont, CA, USA

- Designed **10+ mechanical assemblies**, machine guards, and access-control systems in SolidWorks for automated manufacturing equipment, producing fabrication-ready CAD models and engineering drawings.
- Performed root-cause analysis on **15+ electro-mechanical failures** across automated manufacturing systems, identifying mechanical, electrical, and process issues to support corrective actions.
- Conducted testing, calibration, and validation on **5+ manufacturing systems**, identifying performance deficiencies and supporting engineering modifications to improve operational reliability.

INVOLVEMENT

Propulsion Lead – Mudd Amateur Rocketry Club

Harvey Mudd College • August 2024 – Present

- Led mechanical development of **Apollyon I**, a full-M-class FAR Unlimited competition rocket, coordinating **40+ members** across propulsion, structures, avionics, recovery, and simulation to achieve **1st Place at FAR Unlimited 2026**.
- Directed **systems integration** and launch-readiness reviews across **5 subteams**, driving design decisions throughout the development cycle.
- Evaluated **10+ configurations** for motor selection, stability margins, recovery sizing, and composite airframe design using OpenRocket, RASAero II, MATLAB, SolidWorks, and COMSOL.
- Manufactured and integrated flight hardware for **Gladius III**, contributing to a **3rd Place finish (2025)** and validating simulations against flight data with **99.95% apogee accuracy**.

Mechanical Lead – Mudd Robo-Sub Team

Harvey Mudd College • August 2024 – Present

- Designed and prototyped **3+ iterations** of a torpedo release mechanism for an autonomous underwater vehicle, improving deployment reliability and repeatability under submerged conditions.
- Performed **FEA in COMSOL** to evaluate stress concentrations and deformation under operational loading across **5+ design configurations**, guiding structural optimization and improving robustness.
- Redesigned PCB layouts to improve signal integrity, reduce electrical noise, and enhance subsystem reliability in underwater environments.

PROJECT

Autonomous Underwater Robot with Multi-Sensor Environmental Monitoring

Harvey Mudd College • drive.google.com/file/d/1G8g21u103cptKQe5Qd1aRN1xacnTTyyP/view?usp=drive_link • January 2026 – May 2026

- Designed and integrated pressure, pH, turbidity, and temperature sensing systems for an autonomous underwater vehicle capable of collecting environmental data to depths of 2 m.
- Developed analog signal-conditioning circuitry and **closed-loop depth control** on a **Teensy 4.0**, integrating MATLAB regression models with Arduino-based feedback control.
- Conducted **field testing** to validate sensor performance and analyze environmental trends across turbidity, pH, temperature, and depth measurements.

Convertible Sofa-Bed System for Camper Van Applications

Harvey Mudd College • drive.google.com/file/d/14agxy6UnMwTgToU1uOJSBQtKUzP2dwkG/view?usp=sharing • January 2025 – May 2025

- Led the end-to-end design of a deployable sofa-bed system for a camper van client, generating **5+ design concepts** while balancing ergonomics, manufacturability, structural performance, and storage constraints.
- Evaluated **3 support-leg architectures** and multiple hinge configurations through prototype testing and FEA-driven optimization to increase stiffness and reduce peak stresses.
- Created CAD models, fabricated a full-scale prototype, and validated system performance through deployment-force testing and full configuration transitions completed in **82 seconds**.

SKILLS

CAD & Design: SolidWorks, Engineering Drawings, GD&T, DFM, Mechanical Design, Materials Selection, Circuit Design, KiCad

Simulation & Analysis: FEA, CFD, MATLAB, Simulink, COMSOL, Data Analysis, Flight Simulation, OpenRocket, RASAero II, Electrical Fault Analysis

Fabrication: Machining, CNC Machining, 3D Printing, Mechanical Assembly, Prototyping, Composite Layups, Laser Cutting, Waterjet Cutting

Programming: Python, MATLAB, C++, Git, LabVIEW, Arduino IDE, Embedded Systems Programming

Certifications: NAR Level 1 High-Power Rocketry